Information spectrum and design dependence of noisy-state distinguishability

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We bound the average trace distance between two independent random pure states, drawn from exact 2-designs, after arbitrary Pauli noise. Finite-size bounds place it between two information-spectrum probabilities of the error surprisal. For i.i.d. errors this gives a sharp threshold at the hashing entropy and, near nondegenerate critical noise, a Gaussian crossover with explicit error. At degenerate critical noise, two exact 2-designs have different limits, and random stabilizer states, a 3-design, do not attain the known decay-rate bound.

Introduction.—Random states and state designs provide tractable models of scrambled quantum information, and noisy random circuits are a standard setting for how noise erases it [1, 2]. Robustness depends on the operational task: distinguishing two known states, preserving an unknown quantum state, and learning a state by restricted measurements are different requirements. We consider unrestricted binary discrimination with equal priors. For a fixed channel $\mathcal{N}$, its ensemble-averaged optimal success probability is $(1+\overline{T})/2$, where

$$\overline{T} = \mathbb{E}_{\psi_1,\psi_2} \frac{1}{2} \|\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2)\|_1. \tag{1}$$

The measurement may depend on the two known candidate states.

Two recent works bound essentially this quantity for tensor-power noise. Ibarrondo and Stilck França [3] study the average contraction coefficient, the ratio of output to input trace distance. For any 1-design they prove it tends to one when the Choi-state entropy satisfies $S(\tau_{\mathcal{N}}) < -\log_2 \|\mathcal{N}(I/d_{\text{in}})\|_\infty$, which reduces to the logarithm of the output dimension for unital channels such as Pauli channels. Their proof projects onto the span of typical Kraus images of the first state and averages the second state's first moment. Combined with a 2-design upper bound, this gives local-depolarizing thresholds near 0.25 and 0.42, leaving the interval between them open. Liu *et al.* [4] derived conditional- and collision-entropy thresholds at the same two values by decoupling, with an $O(n^{-1/2})$ upper bound in between. For random codes under depolarizing noise, Sommers *et al.* [5] identified the hashing transition. They argued for an $n^{-1/2}$ window in the coherent information, at fixed logical size and vanishing rate, from spectral analysis and numerics. Information-spectrum bounds, Rényi exponents, and Gaussian second-order rates are established tools in privacy amplification and quantum soft covering [6–8].

Our contribution is narrower and specific to Pauli noise. We match the projector lower bound of Ref. [3], specialized to Pauli errors, with a second-moment upper bound on the same information-spectrum quantity. The result is a finite-size bracket valid for correlated Pauli distributions and exact 2-designs. For i.i.d. errors it places the threshold exactly at the hashing entropy and yields the Gaussian crossover profile, with an explicit error, for nondegenerate critical channels. For correlated errors the bounds depend on the full information spectrum; the Shannon entropy alone need not determine the limiting distinguishability. Exponential decay throughout the formerly open interval already follows from existing smoothed decoupling bounds; we record this rather than claim it.

Both prior analyses state their conclusions for an arbitrary design of the relevant order [3, 4], and random stabilizer states are a common stand-in for Haar states because they form 3-designs [9]. We test how far such design-level statements reach. Our bounds hold for every exact 2-design, but the critical limit does not: a degenerate critical channel separates two exact 2-designs. Away from threshold, a bound for random stabilizer states, which form a 3-design, shows that they do not attain the known achievable decay rate. Neither the bounds nor our numerics determine the exact decay rate at general noise strength.

Finite-size bounds.—Let $d = 2^n$, $\mathcal{P}_n = \{I, X, Y, Z\}^{\otimes n}$, and

$$\mathcal{N}_p(\rho) = \sum_{P\in\mathcal{P}_n} p_P P\rho P, \qquad s(P) = -\log_2 p_P, \tag{2}$$

where s is sampled with $P \sim p$ on the support of p. All logarithms and entropies below use base two. The channel is fixed independently of the state samples. Each $\psi_i = |\psi_i\rangle\langle\psi_i|$ is sampled independently from an exact pure-state 2-design on the *full* d-dimensional Hilbert space:

$$\mathbb{E}\psi_i = I/d, \qquad \mathbb{E}\psi_i^{\otimes 2} = (I+F)/[d(d+1)], \tag{3}$$

where F swaps the two copies. The two ensembles need not be identical. Examples include Haar states and uniformly random qubit stabilizer states [9].

Theorem 1. *For every Pauli distribution p and $a \geq 0$,*

$$\mathbb{P}[s < n-a] - 2^{-a} \leq \overline{T} \leq \mathbb{P}[s < n+a] + 2^{-a/2}/\sqrt{2}. \tag{4}$$

More precisely, for any subset $\mathcal{G} \subseteq \mathcal{P}_n$ and any partition

$\{\mathcal{C}_j\}$, *put* $m_j = \sum_{\mathcal{C}_j} p_P$ *and* $v_j = \sum_{\mathcal{C}_j} p_P^2$. *Then*

$$\overline{T} \geq p(\mathcal{G}) - |\mathcal{G}|/d, \tag{5}$$

$$\overline{T} \leq \sum_j \min\left\{ m_j, \sqrt{\frac{d^2 v_j - m_j^2}{2(d+1)}} \right\}. \tag{6}$$

Proof. Fix ψ_1 and project onto the span of $\{P|\psi_1\rangle : P \in \mathcal{G}\}$, with projector Π. For each realization, the variational formula for trace distance gives $\frac{1}{2}\|\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)\|_1 \geq \operatorname{tr}\Pi[\mathcal{N}_p(\psi_1) - \mathcal{N}_p(\psi_2)]$; we then average over ψ_2 and ψ_1. The first term is at least $p(\mathcal{G})$; independence, the first moment of ψ_2, and unitality give conditional mean $\operatorname{tr}\Pi/d$ for the second. Since $\operatorname{tr}\Pi \leq |\mathcal{G}|$, this proves (5). Only the second state's 1-design property is needed here; the first can be any fixed pure state. This is the typical-Kraus-image argument of Ref. [3] (Theorem 18 there), applied to an arbitrary set of Pauli errors instead of a typical set of an i.i.d. Kraus decomposition.

For the upper bound let $\Delta = \psi_1 - \psi_2$ and $\mathcal{N}_j(\rho) = \sum_{\mathcal{C}_j} p_P P\rho P$. Pauli orthogonality and (3) imply

$$\begin{aligned} \mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_i)^2 &= \frac{dv_j + m_j^2}{d+1}, \\ \mathbb{E}\operatorname{tr}\mathcal{N}_j(\psi_1)\mathcal{N}_j(\psi_2) &= m_j^2/d, \\ \mathbb{E}\operatorname{tr}\mathcal{N}_j(\Delta)^2 &= \frac{2d}{d+1}v_j - \frac{2m_j^2}{d(d+1)}. \end{aligned} \tag{7}$$

Jensen's inequality and $\|A\|_1 \leq \sqrt{d}\|A\|_2$ give the square-root term in (6); the triangle inequality also gives $\frac{1}{2}\|\mathcal{N}_j(\Delta)\|_1 \leq m_j$. Sum over classes.

For the lower side of (4), use $\mathcal{G} = \{s < n - a\}$, which has fewer than $d2^{-a}$ elements. For the upper side split at $s = n + a$. Bound the head by its mass and the tail by $\sqrt{dv/2}$. On the tail $p_P \leq 2^{-n-a}$, so $v \leq 2^{-n-a}$. □

The two CDF arguments in (4) differ. Thus this is a *bracketing inequality*, not a finite-size identity with $\mathbb{P}[s < n]$. Concentration and regularity of s make the bracket informative. The event $s < n$ means $p_P > 1/d$; it is not synonymous with selecting the d most likely errors. The subspace test detects a collection of error images and does not identify individual errors or construct a quantum recovery map. The lower bound is informative only when some single error has probability above $1/d$. For i.i.d. noise this requires $\max_i p_i > 1/2$, i.e., $r < 2/3$ for depolarizing noise; beyond that every choice of $\mathcal{G}$ gives a nonpositive bound [Fig. 1(a)].

For correlated noise the bracket remains valid, but the Shannon entropy need not locate the threshold; in the following example the bracket still gives the correct limit. Global depolarization $\mathcal{N}(\rho) = \theta\rho + (1-\theta)I/d$, $0 < \theta < 1$, is a Pauli channel with $p_I = \theta + (1-\theta)/d^2$ and $p_P = (1-\theta)/d^2$ otherwise. Its entropy rate tends to $2(1-\theta)$, which exceeds one for $\theta < 1/2$. Yet $\mathcal{N}(\psi_1) - \mathcal{N}(\psi_2) = \theta(\psi_1 - \psi_2)$, so $\overline{T} \to \theta$. The bracket (4) with $a = n/2 - 1$ reproduces this: only the identity has surprisal below $n - a$, and every other error lies above $n + a$, so $|\overline{T} - \theta| \leq 2^{-n/4+1}$ for large n.

I.i.d. noise and the critical window.—Now let $p = \boldsymbol{p}^{\otimes n}$ for a single-qubit distribution $\boldsymbol{p} = (p_0, p_x, p_y, p_z)$. Write

$$H = \mathbb{E}[-\log_2 p_i], \qquad \sigma^2 = \operatorname{Var}[-\log_2 p_i]. \tag{8}$$

For fixed $\boldsymbol{p}$ with $H < 1$, choose $a = \epsilon n$ with $0 < \epsilon < 1 - H$ in the lower bound. A finite-alphabet large-deviation bound [10] gives $\overline{T} \geq 1 - e^{-\Omega(n)}$. The upper bound similarly gives $\overline{T} \leq e^{-\Omega(n)}$ if $H > 1$. The constants depend on the fixed channel. For a qubit Pauli channel $1 - H$ is the coherent information at maximally mixed input [11, 12]. Its zero locates the threshold for this observable, not a general quantum-capacity boundary: degenerate codes transmit quantum information beyond the hashing point [13]. For unital Pauli noise, the near-unity regime $H < 1$ coincides with the Choi-entropy condition of Ref. [3]; the matching upper bound shows that this condition is also where distinguishability is lost.

Exponential decay for fixed $H > 1$ also follows from existing results. It follows directly from the unsmoothed Rényi bound (16) below, by choosing $\alpha > 1$ close enough to one that $H_\alpha(\boldsymbol{p}) > 1$. It also follows from the smoothed decoupling bound of Ref. [4], $\overline{T} \leq 2^{-H_2^\delta/2} + 4\delta$. There H_2^δ is the smoothed sandwiched conditional collision entropy of the Choi state, conditioned on its own marginal, and Lemma 1 of Ref. [4] states an AEP for it directly: $H_2^\delta \geq nh - C\sqrt{n\log_2(2/\delta^2)}$ with $h = H - 1 > 0$ and a channel-dependent constant C. Choosing $\delta = 2^{-\kappa n}$ with sufficiently small fixed $\kappa > 0$ makes both terms exponentially small. (That lemma is derived from the AEP for the smooth min-entropy [14] through $H_{\min} \leq H_2$, which is Eq. (48) of Ref. [15] at $\alpha \to \infty$. Ref. [4] smooths over normalized states in purified distance, whereas Ref. [14] uses subnormalized states, so the smoothing parameters of the two need not coincide. The unsmoothed route above avoids this issue.) The AEP condition $n \geq (8/5)\log_2(2/\delta^2)$ of Ref. [14] permits $\kappa < 5/16$ for large n. We therefore do not regard exclusion of an algebraic region at fixed $H > 1$ as a new decay mechanism.

For the crossover, allow the single-qubit distribution $\boldsymbol{p}_n$ to vary with n; errors remain i.i.d. within each n-qubit system. Let H_n, σ_n be its entropy and standard deviation and $x_n = \sqrt{n}(1 - H_n)/\sigma_n$.

Corollary 1. *Suppose* $\boldsymbol{p}_n \to \boldsymbol{p}_*$, $H(\boldsymbol{p}_*) = 1$, *and* $\sigma(\boldsymbol{p}_*) > 0$. *Uniformly over independent exact pure-state 2-design inputs and i.i.d. errors,*

$$\overline{T}_n = \Phi(x_n) + O\left(\frac{\log n}{\sqrt{n}}\right), \tag{9}$$

where Φ *is the standard normal CDF. In particular, if* $x_n \to x$, *then* $\overline{T}_n \to \Phi(x)$.

Proof. Let $\rho_n = \mathbb{E}|-\log_2 p_{n,i} - H_n|^3$ and $b_n = C_{\mathrm{BE}}\rho_n/(\sigma_n^3\sqrt{n})$. The Berry–Esseen theorem [16] applied to the error surprisal gives

$$\overline{T}_n \geq \Phi\left(x_n - \frac{a}{\sqrt{n}\sigma_n}\right) - b_n - 2^{-a},$$
$$\overline{T}_n \leq \Phi\left(x_n + \frac{a}{\sqrt{n}\sigma_n}\right) + b_n + 2^{-a/2}/\sqrt{2}. \tag{10}$$

The four-letter alphabet bounds the third moment uniformly, since $t|\log t|^3 \to 0$ at zero, and σ_n is bounded away from zero. Hence $b_n = O(n^{-1/2})$. Set $a = 2\log_2 n$ and use the bounded density of Φ to obtain (9). □

For $\mathcal{N}_r(\rho) = (1-r)\rho + rI/2$, $0 \leq r \leq 1$, put $q = 3r/4$. Then $\boldsymbol{p} = (1-q, q/3, q/3, q/3)$ and $H = h_2(q) + q\log_2 3$. The nondegenerate critical point is $q_c = 0.189290\ldots$, or $r_c = 0.252386\ldots$. A smooth transverse crossing of $H = 1$ therefore has an $n^{-1/2}$ noise window. Equation (9) specifies its pairwise trace-distance profile; it does not assert a common finite-size crossing of all curves. Figure 1 separates small-system state-vector sampling from large-system evaluation of rigorous bounds.

A degenerate critical point.—The variance assumption cannot be omitted. Let $\boldsymbol{p}_* = (1/2, 0, 0, 1/2)$; the channel is complete dephasing in the computational basis, with $H = 1$ and $\sigma = 0$. In prime-power dimension $d = 2^n$, the uniform union of a complete set of $d+1$ mutually unbiased bases (MUBs) is an exact pure-state 2-design [17]. Choose the computational basis as one member. Its d states remain unchanged by dephasing; every state from the other bases maps to I/d. Counting pairs gives

$$\overline{T}_{\mathrm{MUB}} = \frac{2d+1}{(d+1)^2}(1-1/d)$$
$$= \frac{(d-1)(2d+1)}{d(d+1)^2} \longrightarrow 0. \tag{11}$$

For Haar states, the outputs are diagonal, so their trace distance equals the total-variation distance of the diagonal probabilities. These probabilities have a Dirichlet$(1,\ldots,1)$ distribution. Two independent coordinates $X, Y \sim \mathrm{Beta}(1, d-1)$ obey $\mathbb{E}\min(X,Y) = \int_0^1 (1-t)^{2d-2}dt = (2d-1)^{-1}$. Consequently,

$$\overline{T}_{\mathrm{Haar}} = \frac{d}{2}\mathbb{E}|X-Y| = \frac{d-1}{2d-1} \longrightarrow \frac{1}{2}. \tag{12}$$

Both ensembles satisfy identical first and second moments, but their critical limits differ (Fig. 2). This example concerns a legitimate Pauli channel at the entropy threshold, not approximate designs. It sharpens the observation of Ref. [3] that computational-basis states, a 1-design, are not contracted by dephasing: here even matching second moments do not fix the critical limit.

Even positive variance at each finite n is insufficient without control of the limiting distribution. For example, set $\epsilon_n = n^{-6}$ and $\boldsymbol{p}_n = (1/2-\epsilon_n+\delta_n, \epsilon_n, \epsilon_n, 1/2-\epsilon_n-\delta_n)$, where $\delta_n > 0$ solves $H(\boldsymbol{p}_n) = 1$. For large n this solution is unique and interior. Binary entropy bounds give $\delta_n = O(\sqrt{\epsilon_n \log(1/\epsilon_n)})$, so $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1 \to 0$. Replacing one tensor factor at a time and using $\|\mathcal{A}\otimes\mathcal{B} - \mathcal{A}'\otimes\mathcal{B}'\|_\diamond \leq \|\mathcal{A}-\mathcal{A}'\|_\diamond + \|\mathcal{B}-\mathcal{B}'\|_\diamond$ for channels, the tensor channels differ in diamond norm by at most $n\|\boldsymbol{p}_n - \boldsymbol{p}_*\|_1$. Thus MUB inputs still have $\overline{T}_n \to 0$, although each $\sigma_n > 0$ and $x_n = 0$. The regularity in Corollary 1 excludes this sequence.

Decay bounds and their scope.—For i.i.d. errors, partition Pauli strings into empirical types t with masses m_t and cardinalities N_t. All strings in a type have equal probability. Relaxing (6) therefore gives

$$\overline{T}_n \leq \sum_t m_t \min\{1, \sqrt{2^n/(2N_t)}\} \leq \mathrm{poly}(n)\, 2^{-nE(\boldsymbol{p})}, \tag{13}$$

where standard type bounds [10] $N_t = 2^{nH(t)+O(\log n)}$ and $m_t \leq 2^{-nD(t\|\boldsymbol{p})}$ yield

$$E(\boldsymbol{p}) = \min_{t \ll \boldsymbol{p}}\{D(t\|\boldsymbol{p}) + \tfrac{1}{2}[H(t)-1]_+\}$$
$$= \max_{0\leq s\leq 1} \frac{s}{1+s}[H_{1+s}(\boldsymbol{p}) - 1]. \tag{14}$$

Here $D(t\|p) = \sum_i t_i \log_2(t_i/p_i)$, $H_\alpha(p) = (1-\alpha)^{-1}\log_2\sum_i p_i^\alpha$, and the $s = 0$ value is defined by continuity. To obtain the second line, write $[H-1]_+/2 = \max_{0\leq u\leq 1/2} u(H-1)$, interchange convex–concave extrema on the support simplex, minimize over t, and set $s = u/(1-u)$. The minimizing distribution at fixed s is proportional to p_i^{1+s}.

Equation (14) is the established universal$_2$ privacy-amplification achievable exponent at rate one [6]. Its collision-entropy branch and escort-distribution critical rate are properties of that optimization. They do not prove a nonanalyticity of the true distance. The same bound follows from decoupling without smoothing [7]: the channel family $\mathcal{R}^\psi(X) = \sum_P X_{PP} P\psi P$ is 1-randomizing. Writing X_P for the operator on a reference system attached to label P, Eq. (3) gives

$$\mathbb{E}_\psi \Big\| \sum_P (P\psi P - I/d)\otimes X_P \Big\|_2^2$$
$$= \frac{d}{d+1}\sum_P \|X_P\|_2^2 - \frac{\|\sum_P X_P\|_2^2}{d(d+1)} \leq \|X\|_2^2. \tag{15}$$

Applying Theorem 8 of Ref. [7] to diagonal p gives, for $1 < \alpha \leq 2$,

$$\mathbb{E}\|\mathcal{N}_p(\psi) - I/d\|_1 \leq 2^{2/\alpha-1} 2^{-\frac{\alpha-1}{\alpha}[H_\alpha(p) - \log_2 d]}. \tag{16}$$

The two-state triangle inequality and i.i.d. additivity recover (14), without a type-count prefactor. No matching converse for the present observable is implied by these references.

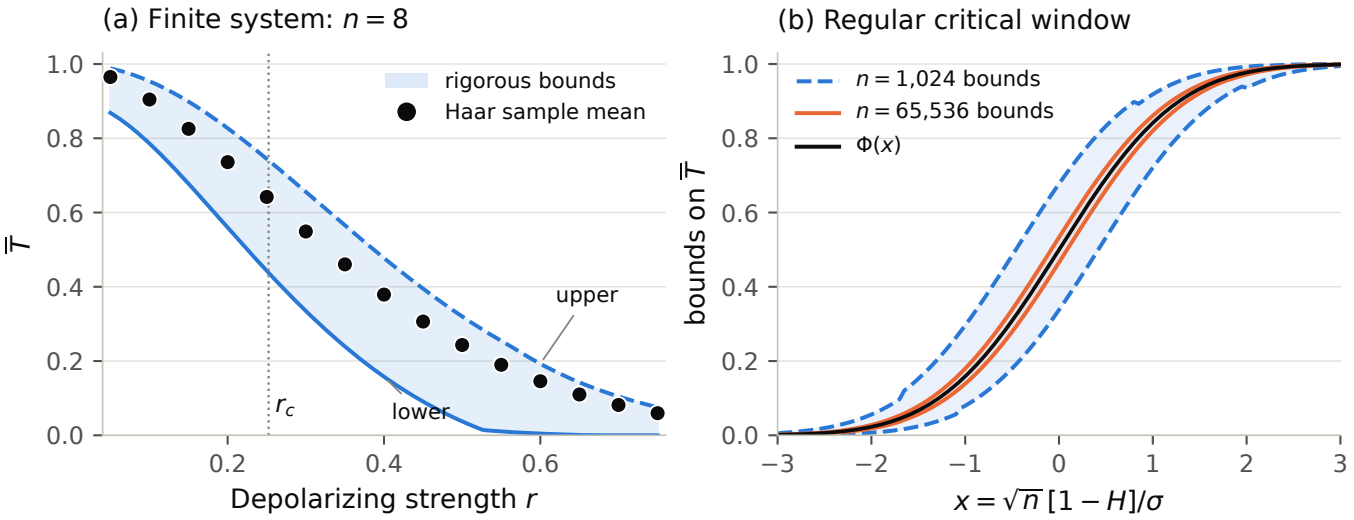


FIG. 1. Finite-size statements for local depolarizing noise. (a) At $n = 8$, the shaded interval lies between the rigorous projector lower bound (solid) and two-class upper bound (dashed) of Eqs. (5)–(6), optimized over error-weight cutoffs; the dotted vertical line marks r_c. The lower bound vanishes for $r \geq 2/3$, where no single error has probability above $1/d$. Points are means from 64 independent Haar-state pairs, using full density matrices; bars are one sample standard error. The same pairs are reused across noise strengths. These samples check finite-size consistency and do not establish an asymptotic rate. (b) The explicit information-spectrum bracket (4), with $a = 2\log_2 n$, is evaluated from binomial CDFs at fixed $x = \sqrt{n}(1 - H)/\sigma$. Shaded intervals for $n = 1024$ and 65536 approach $\Phi(x)$ (black). These are bounds, not simulated distances.

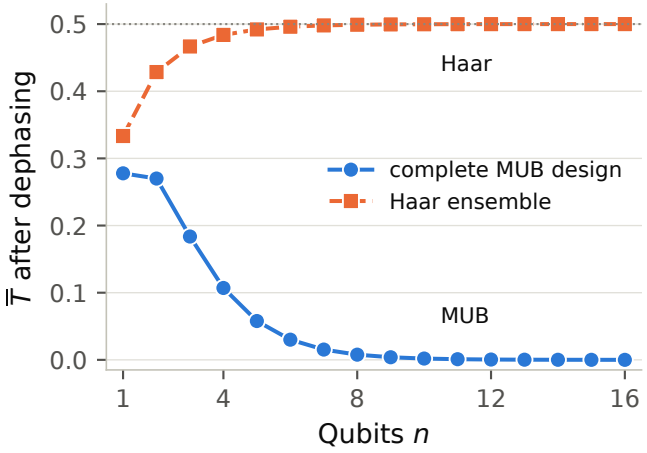


FIG. 2. An exact counterexample to design-independent critical behavior. Under complete dephasing, both ensembles are pure-state 2-designs and the error entropy is one bit per qubit. Equations (11) and (12), evaluated without sampling, give different limits. The dashed gray reference marks 1/2.

A limited matching statement does follow near the depolarizing threshold. For any distribution $t \ll p$ with $c(t,p) = D(t\|p) + H(t) < 1$, Eq. (5), applied to nearby empirical types, gives $\overline{T}_n \geq 2^{-n[D(t\|p)+o(1)]}$: its mass exceeds its dimension cost by the factor $1 - 2^{-n[1-c(t,p)]}$. Thus, writing $R_- = \liminf_n[-n^{-1}\log_2 \overline{T}_n]$ and R_+ for the corresponding limsup,

$$E(p) \leq R_- \leq R_+ \leq E_{\mathrm{proj}}(p),$$
$$E_{\mathrm{proj}}(p) = \inf_{c(t,p)<1} D(t\|p), \tag{17}$$

with an empty infimum interpreted as $+\infty$. For depolarizing $q = 3r/4$ just above q_c, symmetry reduces the infimum to $t = (1-f, f/3, f/3, f/3)$. Its boundary $f = f_0(q)$ satisfies $-(1-f)\log_2(1-q) - f\log_2(q/3) = 1$. Here $f_0(q_c) = q_c$ and $f_0'(q_c) = 0$, whereas the minimizer in (14) has $f = q_c$ near threshold. Expanding the binary relative entropies on both sides therefore yields

$$R_\pm(r) = \kappa(r - r_c)^2 + O((r - r_c)^3),$$
$$\kappa = \frac{9}{32\ln 2\, q_c(1 - q_c)} = 2.64\ldots. \tag{18}$$

The limits in n are taken first, at fixed $r > r_c$. This establishes only the leading near-threshold rate, without assuming a full rate limit exists.

The rate (14) cannot be universally tight over 2-designs. For uniformly random stabilizer states, write $\psi = d^{-1}[I + \sum_{P\neq I} z_P P]$ with $z_P \in \{0, \pm 1\}$ and $\mathbb{E}|z_P| = 1/(d+1)$. Local depolarization multiplies a weight-w coefficient by λ^w, where $\lambda = 1 - r$. Applying the triangle inequality termwise to the difference of two states gives

$$\overline{T}_n^{\mathrm{stab}} \leq \frac{\sum_{P\neq I} \lambda^{w(P)}}{d+1} = \frac{(1+3\lambda)^n - 1}{2^n + 1}. \tag{19}$$

Writing the asymptotic rate as $\liminf_n[-n^{-1}\log_2 \overline{T}_n]$, Eq. (19) at $r = 0.9$ implies a lower bound $1 - \log_2(1.3) =$

$0.621488\ldots$, strictly greater than $E = \frac{1}{2}[1-\log_2(1.03)] = 0.478678\ldots$. This comparison excludes universal equality with (14). Since stabilizer states form a 3-design [9], agreement of the first three moments with Haar does not by itself make the achievable rate tight. It does not determine the exact Haar or stabilizer rate, and it does not show that these rates differ. At $r = 1$ both outputs are exactly I/d and $\overline{T}_n = 0$ for every n.

Discussion.—Pauli error probabilities and Hilbert-space dimension suffice to bound pairwise distinguishability for all exact pure-state 2-designs, through the information spectrum of the error distribution. For i.i.d. errors the matched bounds locate the threshold at the hashing entropy; for correlated errors the entropy alone need not. The bracket gives the Gaussian critical profile when surprisal has nonzero limiting variance. At a degenerate critical channel the answer depends on the ensemble despite identical second moments. Random stabilizer states, a 3-design, do not attain the achievable exponent, and the present bounds do not determine general exact rates. The ingredients—a typical-image projector, second moments, and information-spectrum truncation—are standard; the contribution is their matched combination for this observable and the resulting delineation of what second moments do and do not fix. Binary discrimination of two known states remains distinct from error recovery and channel capacity. Extensions to approximate designs or non-Pauli channels require additional error control; they are not assumed here. Determining ensemble-specific rates away from the transition remains open in this analysis.

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Data availability.—The data that support the findings of this article and the code that generates every figure and numerical check accompany this submission and will be deposited in a public repository [18].

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